

TrES-3b

R-2775

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140630 · created 2026-09-03T23:29:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456838.7754 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.079 +/- 0.053
Transit depth (Rp/Rs)^2	0.62 +/- 0.84 [%]
Orbital Inclination (inc)	79.36 +/- 3.01
Airmass coefficient 1 (a1)	1.235 +/- 0.055
Airmass coefficient 2 (a2)	-0.21 +/- 0.26
Scatter in the residuals of the lightcurve fit is	4.48 %
Best Comparison Star	#2 - [519, 145]
Optimal Aperture	2.9
Optimal Annulus	7.93
Transit Duration (day)	0.019 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.079 $\pm$ 0.053 vs 0.16309 (deviation 1.58 σ)
Duration fit vs published	0.019 d vs 0.0567 d (deviation 1.5 σ)
Mid-time O-C	-0.8 min
Depth SNR	1.5
Residual scatter	4.48 %

Light curve

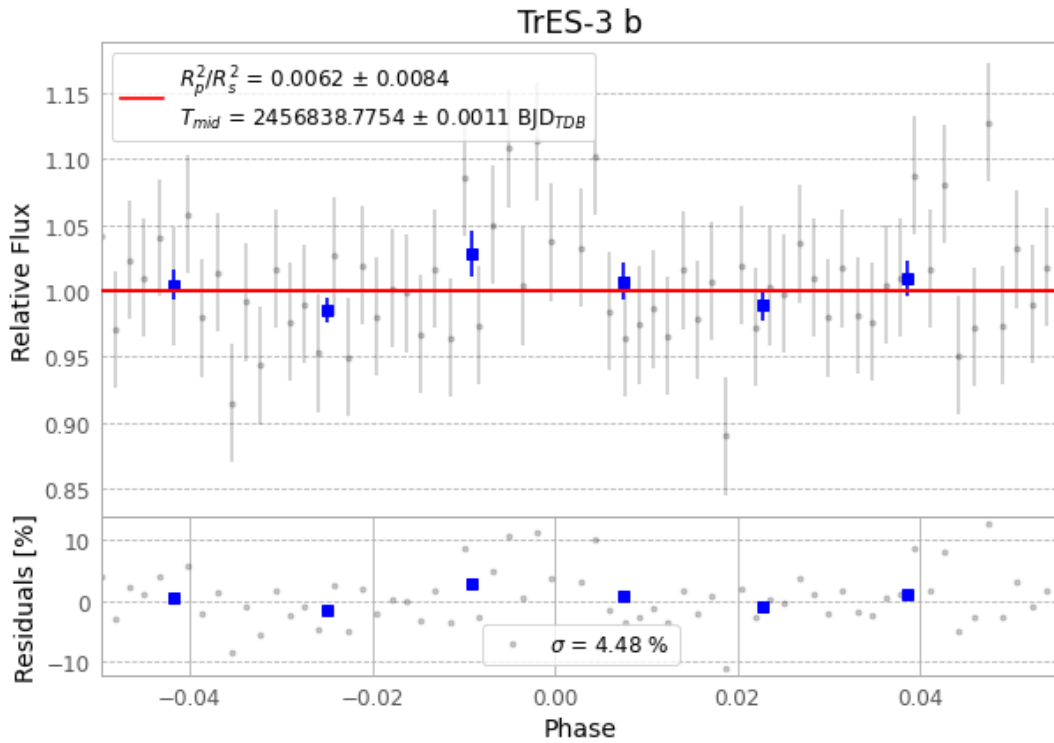


Figure 1 — FinalLightCurve_TrES-3 b_2014-06-29.png (SHA-256 37103cd8facf98b4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0a9c8cc7cacfb617...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3140630045714.FITS → TRES-3140630081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140630.log (3352c76e46e5...), FinalLightCurve_TrES-3 b_2014-06-29.png (37103cd8facf...), FinalLightCurve_TrES-3 b_2014-06-29.csv (b91d7fb40da6...)

Compute resources

Wall time 136.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:29Z.